

# Load Interconnection Request System Impact Study Report

## For Queue Project Q19-1332

Ellettsville 138 kV  
March 27, 2021

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1332. The Interconnection Customer proposes a 184.8 MW large load (data center) facility interconnecting at the Ellettsville 138 kV substation in the Maryland area. The request was received on November 07, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 444 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,718,000.

### **3. Network Impacts**

#### **3.1 Load Deliverability Analysis**

RIO BLANCO 2702-ALLIANCE 138kV: 142.6% loading (247 MVA) following the outage of the SIERRA PACIFIC LUMBER SUBSTATION 138 kV/1 kV Transformer; DFAX = 30.5%.

For the loss of SIERRA PACIFIC LUMBER 138-1 kV Transformer, the HOLOPAW-BRUCE CROSSING 138kV loads to 137.9% of its 290 MVA rating. Project DFAX: 30.5%.

For the loss of SIERRA PACIFIC LUMBER 345 kV/1 kV Transformer Bank, the MCCALL SUBSTATION-PEBBLEDALE 138kV line loads to 130.9% of its 422 MVA rating. Project DFAX: 35.7%.

STILLWATER HYDRO ELECTRIC PROJECT-SANDARIO 138kV: 128.4% loading (232 MVA) following the outage of the SIERRA PACIFIC LUMBER 138-1 kV Auto; DFAX = 19.8%.

The TABER-PEBBLEDALE 138kV is overloaded to 128.3% of its 466 MVA emergency rating for the loss of the SIERRA PACIFIC LUMBER 345-1 kV Auto. The project contributes 35.7% DFAX (66.0 MW) to this overload.

BRUCE CROSSING-RIO BLANCO 2702 138kV: 128.1% loading (294 MVA) following the outage of the SIERRA PACIFIC LUMBER 138-1 kV TX; DFAX = 30.5%.

The STILLWATER HYDRO ELECTRIC PROJECT-MCCALL SUBSTATION 138kV is overloaded to 121.3% of its 261 MVA emergency rating for the loss of the SIERRA PACIFIC LUMBER 138/1kV Transformer Bank. The project contributes 19.8% DFAX (36.6 MW) to this overload.

The GLEN BURNIE-SANDARIO 138kV is overloaded to 109.3% of its 243 MVA emergency rating for the loss of the LAKE CREEK-SIERRA PACIFIC LUMBER SUBSTATION 138kV. The project contributes 22.4% DFAX (41.4 MW) to this overload.

For the loss of HOOKDALE-DIXLETA 138kV Line, the DIXLETA-HOOKDALE 138kV Circuit 3 loads to 107.6% of its 290 MVA rating. Project DFAX: 7.0%.

PAJUILL TC-SN LORENZO 3302 138kV line: 106.7% loading (244 MVA) following the outage of the HOOKDALE-TERRY 138kV; DFAX = 13.4%.

PORT ROBSON-PAJUILL TC 138kV: 106.5% loading (343 MVA) following the outage of the TABER SUBSTATION 345/138 kV XFMR; DFAX = 20.2%.

For the loss of SIERRA PACIFIC LUMBER 138-1 kV Transformer, the MCCALL SUBSTATION-FOOTHILL 138kV Line loads to 105.1% of its 194 MVA rating. Project DFAX: 15.9%.

MOUNT WASHINGTON-LONE PEAK 138kV: 102.0% loading (204 MVA) following the outage of the CIPRES-THOMPSON PEAK TAP 138kV; DFAX = 6.6%.

For the loss of HOOKDALE-DIXLETA 138kV #1, the HOOKDALE-DIXLETA 138kV Cir. 2 loads to 101.6% of its 307 MVA rating. Project DFAX: 7.0%.

The TERRY-HOOKDALE SUBSTATION 138kV L is overloaded to 101.3% of its 244 MVA emergency rating for the loss of the SAN LORENZO 3302-PAJUIL TC 138kV line. The project contributes 10.1% DFAX (18.7 MW) to this overload.

The HOLOPAW-PORT ROBSON 138kV line is overloaded to 100.9% of its 334 MVA emergency rating for the loss of the PEBBLEDAL SUBSTATION-TABER 138kV. The project contributes 21.4% DFAX (39.6 MW) to this overload.

WIGWAM-UPSCO 345kV: 100.6% loading (1,509 MVA) following the outage of the MORGANVILLE-WESTWOOD GENERATION LLC 345kV Line; DFAX = 5.1%.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                    | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| ALLIANCE-RIO BLANCO 2702 138kV                              | Rebuild line with higher-capacity conductor | \$1,042,000   | 48% (\$496,000)       |
| HOLOPAW-BRUCE COSSING 138kV                                 | Rebuild line with higher-capacity conductor | \$3,655,000   | 48% (\$1,740,000)     |
| PEBBLEDAL-MCCALL SUBSTATION 138kV                           | Rebuild line with higher-capacity conductor | \$1,613,000   | 42% (\$683,000)       |
| SANDARIO-STILLWATER HYDRO ELECTRIC PROJECT SUBSTATION 138kV | Rebuild line with higher-capacity conductor | \$995,000     | 60% (\$599,000)       |
| TABER-PEBBLEDAL 138kV                                       | Rebuild line with higher-capacity conductor | \$2,822,000   | 42% (\$1,195,000)     |
| RIO BLANCO 2702-BRUCE CROSSING 138kV                        | Rebuild line with higher-capacity conductor | \$2,109,000   | 48% (\$1,004,000)     |
| STILLWATER HYDRO ELECTRIC PROJECT-MCCALL SUBSTATION 138kV   | Rebuild line with higher-capacity conductor | \$2,678,000   | 60% (\$1,614,000)     |
| GLEN BURNIE-SANDARIO 138kV                                  | Reconductor line                            | \$917,000     | 58% (\$531,000)       |
| HOOKDALE-DIXLETA SUBSTATION 138kV Line 3                    | Reconductor line                            | \$749,000     | 38% (\$285,000)       |
| PAJUIL TC-SAN LORENZO 3302 138kV                            | Reconductor line                            | \$619,000     | 48% (\$299,000)       |
| PORT ROBSON-PAJUIL TC 138kV                                 | Reconductor line                            | \$1,220,000   | 48% (\$580,000)       |
| FOOTHILL-MCCALL SUBSTATION 138kV                            | Reconductor line                            | \$497,000     | 32% (\$161,000)       |
| MOUNT WASHINGTON-LONE PEAK 138kV                            | Reconductor line                            | \$861,000     | 9% (\$80,000)         |

| Facility                             | Required Upgrade           | Cost Estimate | Allocation to Project |
|--------------------------------------|----------------------------|---------------|-----------------------|
| DIXLETA-HOOKDALE 138kV Line 2        | Reconductor line           | \$835,000     | 38% (\$318,000)       |
| TERRY-HOOKDALE SUBSTATION<br>138kV L | Reconductor line           | \$945,000     | 48% (\$456,000)       |
| PORT ROBSON-HOLPAW 138kV             | Reconductor line           | \$877,000     | 45% (\$393,000)       |
| UPSCO-WIGWAM 345kV                   | Reconductor line           | \$19,851,000  | 1% (\$284,000)        |
|                                      | Total allocated to project |               | \$10,718,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.